

EC400: Exam

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Consider a firm with an optimal (log) price $\bar{p}$, so that each time the actual price p_t set by the firm deviates from this level, the profits go down by $(p_t - \bar{p})^2$. Although $\bar{p}$ maximizes static profits, it might be suboptimal to set the price at this level because each time the firm updates its price, it has to pay adjustment (menu) costs $\kappa\pi_t^2$. Thus, the objective of the firm is to minimize the sum of static losses and adjustment costs:

$$\sum_{t=1}^{\infty} \beta^t \left[(p_t - \bar{p})^2 + \kappa\pi_t^2 \right],$$

subject to the law of motion

$$p_t = p_{t-1} + \pi_t$$

and taking as given the initial price level p_0 .

1. (10 marks) Write down the Lagrangian and take the first-order conditions.
2. (10 marks) Derive the intertemporal optimality condition (Euler equation). Write down the transversality condition?
3. (12 marks) Derive the isolines and solve for all steady states.
4. (14 marks) Which variables are states and which ones are controls in this problem? Draw the phase diagram. Is the steady state a source, a sink, or a saddle point? Is the Blanchard-Kahn principle satisfied? (No analytical proofs are required, the figure with arrows is sufficient to answer these questions)
5. (10 marks) Write down the Bellman equation in terms of the lagged price level $p = p_{t-1}$ and inflation $\pi = \pi_t$.

6. (12 marks) Conjecture that $V(p) = \alpha(p - \bar{p})^2$, where α is an unknown constant. Solve the Bellman equation, i.e. find the value of α and the policy function $\pi(p)$.
7. (8 marks) Discuss which of the variables p_t and π_t are stocks and flows. Rewrite the law of motion for prices/inflation for an arbitrary length of period Δ and take the continuous-time limit.
8. (10 marks) Define discount factor $\rho = -\log \beta$. Write down the Hamiltonian (in terms of p_t, π_t, λ_t) and the optimality conditions according to the maximum principle.
9. (14 marks) Conjecture that $\pi_t = \gamma(p_t - \bar{p})$ and show that the optimality conditions are satisfied. Find the value of γ .